CP APPROVED / EXECUTING

Server Modernization

Replacing aging HPE Gen10 infrastructure with high-performance Dell PowerEdge 17G.

The Imperative

The department's aging server infrastructure, primarily HPE DL560 Gen10, is reaching end-of-support. This transition provides significant benefits for the Real Time Crime Center (RTCC), Domain Awareness System (DAS), and overall virtualization.

Hardware Transition

Decommissioning

- 340x HP ProLiant DL560 Gen10
- 142x HP ProLiant DL580 Gen9
- 54x HP ProLiant DL380 Gen10
- 10x HP ProLiant DL580 Gen8

Procuring

- 387x Dell PowerEdge 17G (vSphere)
- 48x Dell PowerEdge R7725 (Bare Metal)

Performance Gains

2.4x CPU Throughput
Accomplished with 2 sockets instead of 4, saving on VMware licensing.

2.2x Memory Speed
RAM jumps from 2933MT/s to 6400MT/s for faster real-time reporting.

4x I/O Performance
Upgrade from PCIe Gen3 to PCIe Gen5.

Financial Summary

Total Project Cost
$21.7M

Capital
$18.6M

Expense (Maint.)
$3.08M

Consolidation Impact

Server Count	-115 Servers
CPU Cores	-11,230 Cores
Target Racks	-49 Net Racks

**Core reduction significantly lowers VMware vSphere licensing fees.*